

Fast & Slow Pointer Technique - Short Conclusion

1. Use when questions involve finding the middle, halves, or cycles in a linked list.
2. Slow Pointer (Tortoise): Moves 1 node at a time.
3. Fast Pointer (Hare): Moves 2 nodes at a time.
4. When fast reaches the end, slow reaches the middle node.
5. Used to find the middle node in $O(n)$ time and $O(1)$ extra space.
6. Used to detect a cycle (loop) in a linked list.
7. Used to find the starting node of a cycle.
8. Used to check whether a linked list is a palindrome.
9. Used to reorder or split a linked list into two halves.
10. Interview Rule: If the problem mentions middle, cycle, half, palindrome, or reorder, think of the Fast & Slow Pointer approach first.